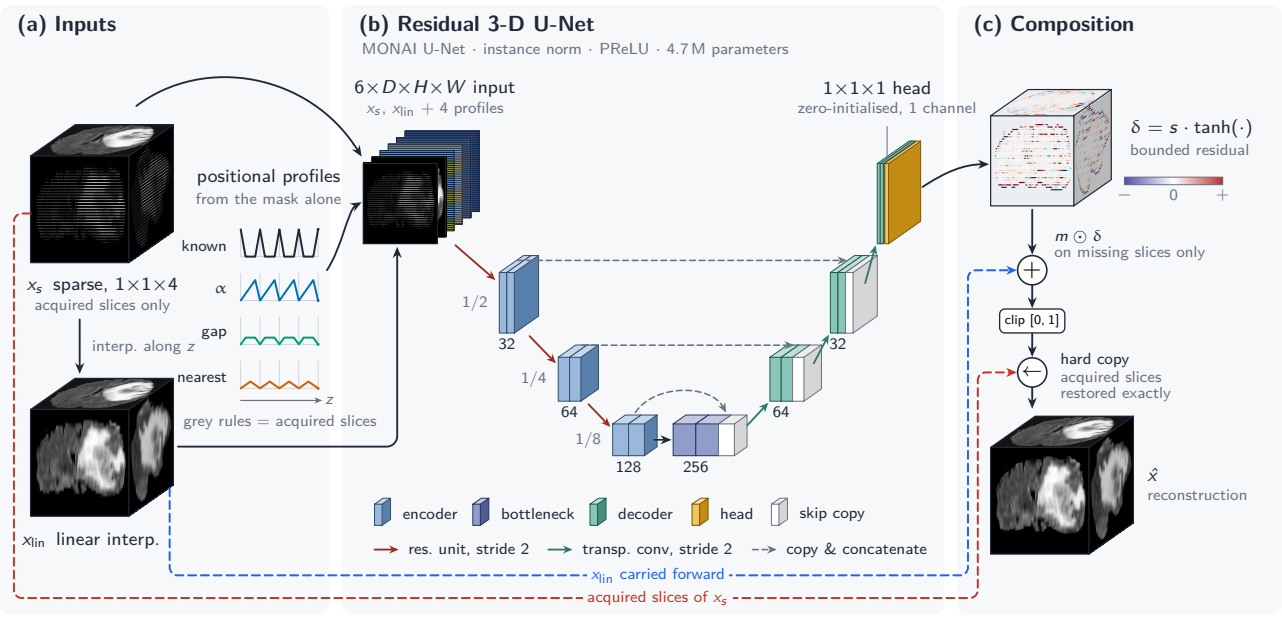

(a) Inputs
positional profiles
from the mask alone
known
α
gap
nearest
z
x_s sparse, 1×1×4
acquired slices only
interp. along z
grey rules = acquired slices
x_{lin} linear interp.
(b) Residual 3-D U-Net
MONAI U-Net · instance norm · PReLU · 4.7 M parameters
6×D×H×W input
x_s, x_{lin} + 4 profiles
1×1×1 head
zero-initialised, 1 channel
1/2
32
1/4
64
1/8
128
256
64
32
encoder
bottleneck
decoder
head
skip copy
res. unit, stride 2
transp. conv, stride 2
copy & concatenate
(c) Composition
$\delta = s \cdot \tanh(\cdot)$
bounded residual
0
$m \odot \delta$
on missing slices only
clip [0, 1]
hard copy
acquired slices
restored exactly
$\hat{x}$
reconstruction
x_{lin} carried forward
acquired slices of x_s